# School of Informatics & Cybersecurity - Assessment Submission Coversheet

## Student Information

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## Module Information

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|---|---|
| Module Title: | Fund of AI & Data Science |
| Module Code: | COMPH2706 |

## Assessment Information

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| Assessment Title: | Data and AI Project |

## Declaration

☐✓ I confirm that I have read and understood the University's Academic Integrity Policy.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/academic-quality-assurance-and-enhancement/academic-integrity/

☐ ✓ I have completed the TU Dublin AI Driving License theory test.

https://www.tudublin.ie/explore/about-the-university/academic-affairs/lta/lta-resources/artificial-intelligence-driving-license/

☐ ✓ I declare that this assessment submission is my own work.

☐ ✓ I confirm that I have read and understood the AI traffic light system used as part of this assessment brief and I have adhered to the level of AI usage allowed for this assessment.

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| **Student signature:** | Eoin Wall |
| **Date:** | 27/04/2026 |

# Public Co lab Link:

https://colab.research.google.com/drive/1vzaeUObZb59mEGMiEtZX4xSDt2L18Bn-?usp=sharing

# Executive introduction:

This document's goal is to show, interpret and understand the given data set, as well as train the data and showcase that data using easy to digest graphs so that people without knowledge of cybersecurity can understand.

The data is a standard cybersecurity database that would be seen in the real world, it consists of different types of data such as protocol, region, failed logins and more technical stuff

In this project, I used 2 different algorithms, decision tree which is a classification model and linear regression which is a predative model, why are these algorithms important to the data? Its because it helps show different types of data which in a real world scenario can help detect and prevent malicious attacks so in terms of cybersecurity this is crucial to keep infrastructure and data safe.

# Introduction:

This document will be showing the data and graphs that I made to understand the data set given and why this data is important for protecting the company in future attacks. I will be documenting my choice of algorithms and what each one is used for, the data I extracted from each model and why this data can be useful in noticing trends and reinforcing existing cybersecurity infrastructure and practices.

By visualizing this data into easier graphs to explain to people not familiar with terminology and understanding of cybersecurity I will be able to explain more clearly about what actions and why they need to be taken to mitigate risk for the company

# Initial dataset:

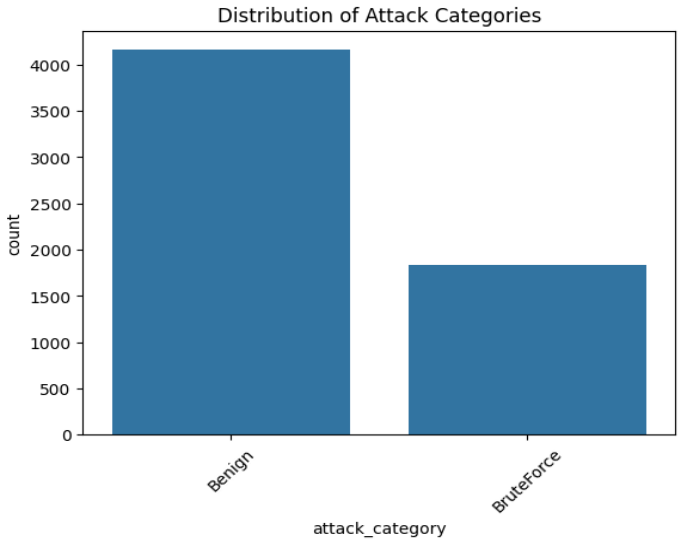
The dataset given contained about 6000 rows of data and 19 columns and as seen in Fig 1, the names of each column, this is the data which the models will be trained off. One of the more important pieces of data that can be seen from the initial dataset is the ratio of "benign" vs "brute force" in the attack category class as seen in Fig 2, this is a very important piece of information as we can see what interactions was harmless such as someone forgetting their password vs deliberate brute force attacks. From what can be seen in the graph between 1500 to 2000 are brute force attacks while the rest are relatively harmless.

Fig 1:

```
  Rows    : 6000
  Columns : 19
```

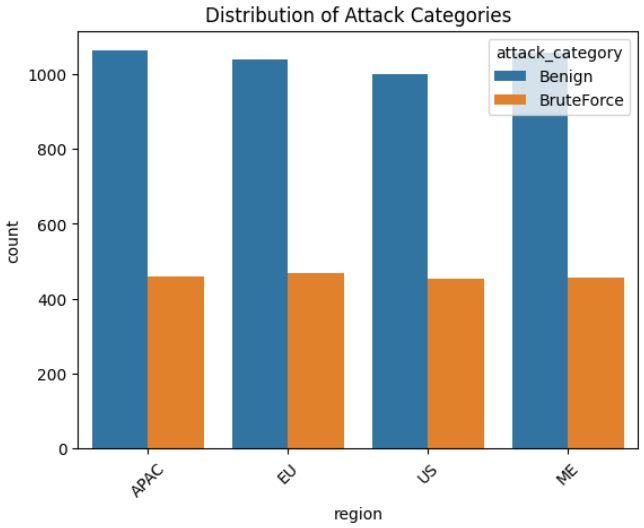
Fig 2:

```
Column names and types:
packet_session_id        str
protocol                 str
device_type              str
vpn_status               str
region                   str
login_attempts         int64
failed_logins          int64
bytes_sent           float64
bytes_received       float64
packet_rate          float64
connection_count       int64
tcp_flags                str
port_number            int64
session_retries        int64
signal_strength      float64
privilege_level          str
anomaly_score        float64
avg_packet_size      float64
attack_category          str
dtype: object
```




Another important graph we can extract from the initial dataset is the region of each attack category as seen in Fig 3, this can tell us that the amount of benign and brute force for each major global region is relatively the same with the US slightly below the other regions, in terms of cybersecurity this means that no one region is causing the majority of brute force attacks, if this wasn't the case then we can isolate or maybe even completely block the one region or at the very least flag any interactions coming from that specific region but that information that this isn't the case is still useful for us.

Fig 3:




Problems with the current data/preprocessing:

Before I started the preprocessing step, I copied the data into a copy, the main reason for this as to not touch the actual data, so any modifications or mistakes don't affect the original data, this helps prevents major accidents or loss of data.

One of the bigger problems with the data is the missing values as I checked for missing values, I found that 3 columns were missing 7% values, those rows being protocol, bytes received and signal strength.

since it was only 7%, I replace the missing values with the mean for the numeric column, bytes received and signal strength, and the mode with protocol. since the data was between 5 to 40% missing the standard procedure was replace it rather than drop the row if it was less than 5% or if it was more than 40% the entire column as that amount of missing data could have skewered results when training the data.

after missing values, I encoded the non-numbered columns of the data set, the reason for this is the 2 algorithms can't accept string, but doing so would make it so graphs are only numbers and no words so to get around that I created another copy called df_model that's only numbers so I feed the models that.

# Features:

I created 3 new features that allowed me to gather more information from the data set. Them being login failing rate, suspicion socre and size anomaly. These new features allow me to display more data and better training for the data algorithms

Model selection:

As stated earlier, I have selected decision tree for my classification and linear regression and ridge regression which is the tuned version of it.

Both models are trained on a 80/20 split which means 80% of it is used to train the model while the last 20% is used to test, if we get the machine full access, it would be the same as given it the answers to a test so we need to leave enough unseen data in order for it to apply what it learns.

For the two models that I decided to go with I chose them based on the data set and what I thought would work best for classification and prediction.

For the first model, decision tree, I chose this one for a few reasons,

1. it is very easy to interpret and has a simple flow chart diagram so it's great for explaining to those not in the cybersecurity sector.
2. Decision trees are non-linear so can be very versatile and can be used on a wide range of different data sets.
3. Can deal with categorical values well

As for the second model, linear regression I also chose that for a few reasons:

1. It's also simple to implement and understand
2. Overfitting can be minimized

Of course, there are disadvantages with both models such as decision tree being prone to overfitting or for linear regression its sensitive to outliers.

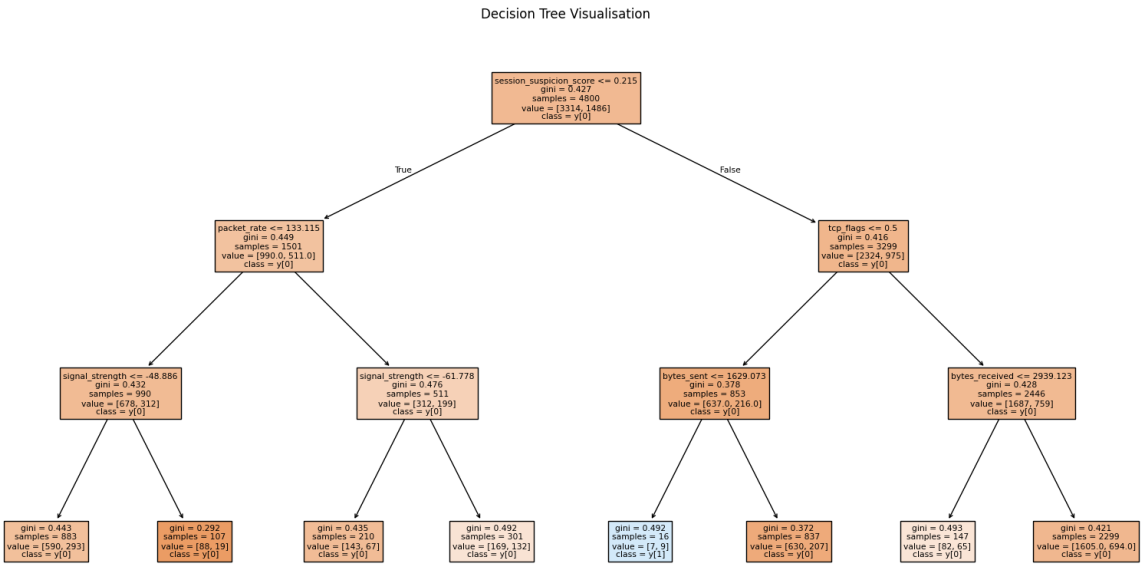
# Results:

Starting with the decision tree, there's a standard and tuned version, but how does a decision tree work? In the basic terms it consists of a "root" node and many branches from it, the machine splits the data set based on features and form groups into those features.

# Decision Tree:

First with the flow chart diagram as seen in Fig 4, this is a visual representation of how the machine is learning and handling the features of the dataset. Unfortunately, this isn't very useful for those who Arnt familiar with cybersecurity.

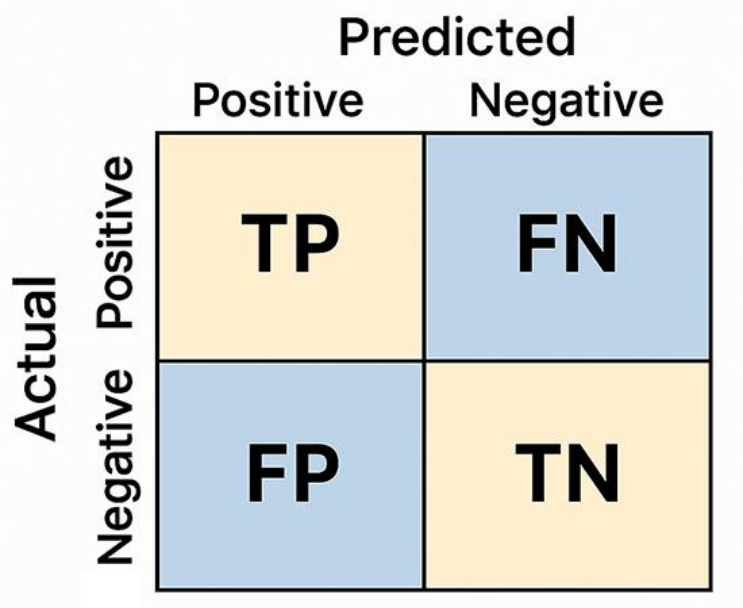
Fig 4:




So, a better graph would be the confusion matrix which is less technical and easier to explain with context on how they function. The top left (True positive) and bottom right (False negative) are the "right" ones meaning that the model was able to correctly predictive the result. The top right (False negative) means the model wrongly predicated a negative case when it was positive, and the bottom left (false positive) is the opposite.

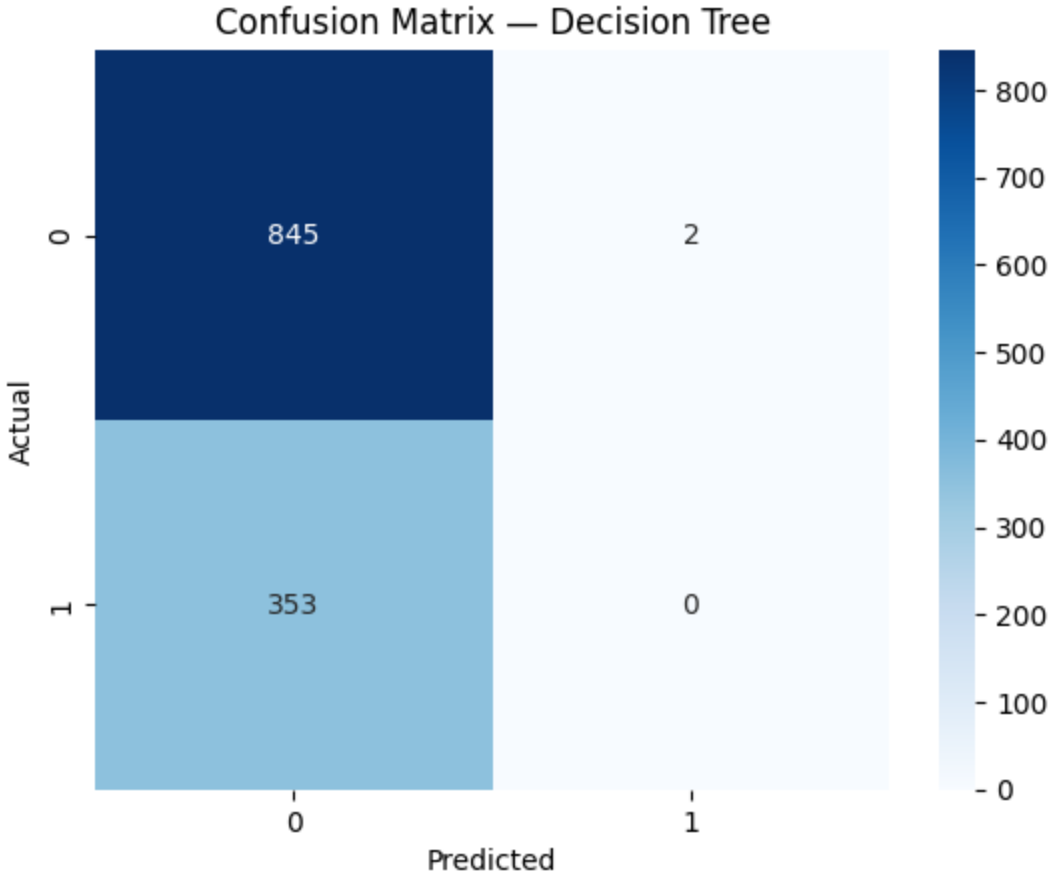
Why is false positive and negative bad? It largely depends on the data, for this business for example, if we got a lot of false positives, it means that we think that our security is working when it is not which is bad for the business and can cost a lot of money

Fig 5:



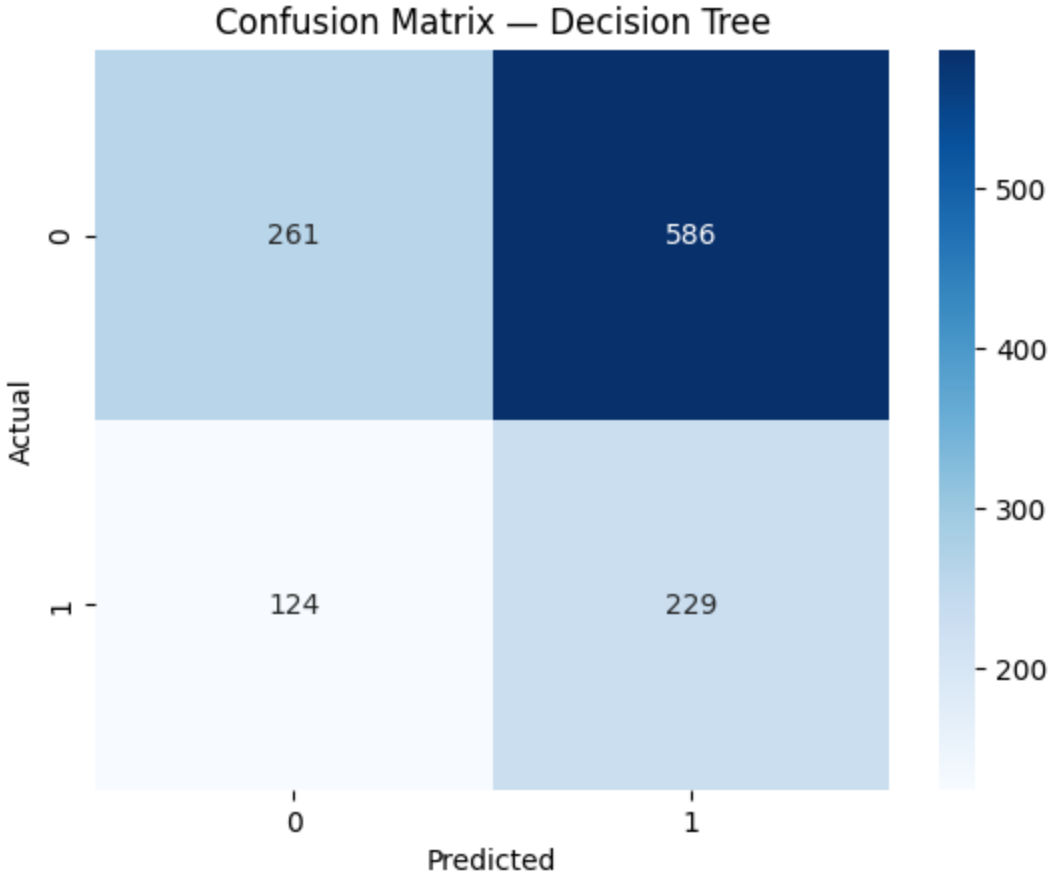

In Fig 6, we can see the confusion matrix of the standard is that there's a lot of true positives this may seem good, but when we look at the other results it means that algorithms are letting too many things through and is not accurate. This is why having a tuned version and standard so you can compare and get the most accurate one.

Fig 6:




Looking at Fig 7, the tuned version of the decision tree we can see the results are a bit more even. This information is vital as it shows that tunning it we can get more accurate results, in the Fig 7 image the false negatives are a lot higher so its not perfect but in any real data set that is expected to happen, it will never be 100% accurate.

Fig 7:




Accuracy is not the only thing we should go off as a lower accuracy might provide better results than an higher accuracy version as seen in Fig 8

Fig 8:

```
Baseline Accuracy:  0.7042
Improved Accuracy:  0.4083
```

If you look at the results, at first glance you would assume that the baseline is more accurate than the improved one. But this isn't the case as a lower accuracy in this case and what we know from the confusion matrix, is that the model is now being more selective in what it chooses.

The best way to see this is in Fig 9 and 10 the standard and tunned/improved one, respectfully, 0 being the non-attacks and 1 being the brute force attacks. In the standard one the precision is higher than the tuned one, but the recall for the attacks is 0, this means its not detecting any of the attacks. But for the tuned one we can see that the recall is 0.58 so its better but an ideal score would be 1 but as this model would be tunned more that score will get closer to a 1.00,

In a business sense this means that if the standard model was employed, nearly all attacks wouldn't be getting flagged which can cause major security issues and lose the company money in GDPR laws and fines.

Fig 9:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.69 | 1.00 | 0.82 | 3314 |
| 1 | 0.43 | 0.00 | 0.01 | 1486 |
| accuracy | | | 0.69 | 4800 |
| macro avg | 0.56 | 0.50 | 0.41 | 4800 |
| weighted avg | 0.61 | 0.69 | 0.57 | 4800 |

Fig 10:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.69 | 0.43 | 0.53 | 3314 |
| 1 | 0.31 | 0.58 | 0.41 | 1486 |
| accuracy | | | 0.47 | 4800 |
| macro avg | 0.50 | 0.50 | 0.47 | 4800 |
| weighted avg | 0.58 | 0.47 | 0.49 | 4800 |

# Linear regression:

The second algorithm was linear regression and ridge for the improved version both of which are a predictive model. In Fig 11, the linear regression line is perfect, while this is very unrealistic but given the small data set it makes sense. What this means is that if this was true, in a business sense it would mean that any prediction the model makes would be 100% accurate.

If the model was more realistic, the dots would be above and below the line which would indicate if the model was overfitting or underfitting. Same as in Fig 12 the graph should not look like that in a different data set.

Another explanation for this could be data leakage that occurred.

Fig 11:

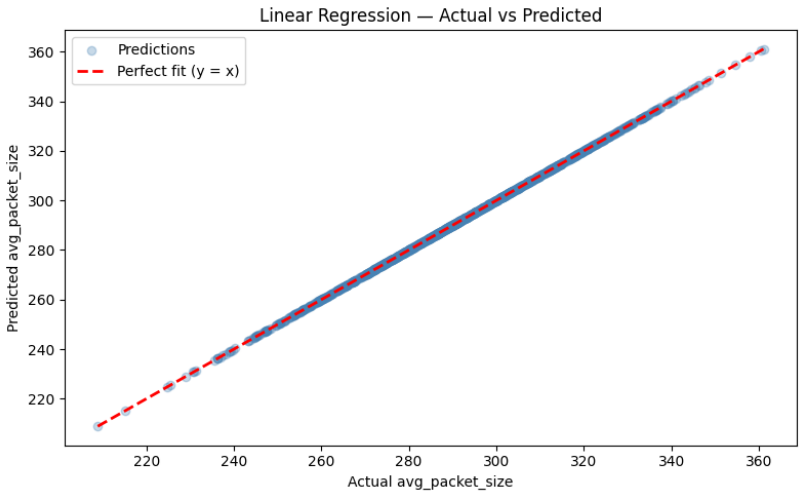
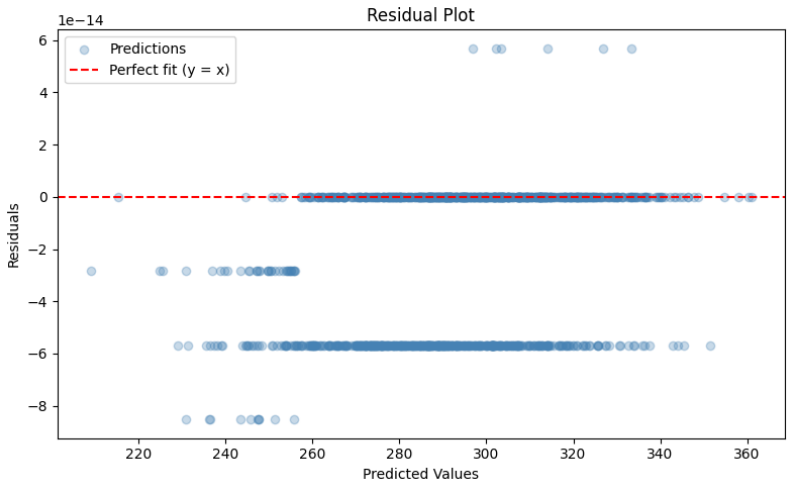



Fig 12:




When we look at the tuned/improved model as seen in Fig 13 and 14, the improved model is different by incorporating “Ridge” which means “is a method of estimating the coefficients of multiple-regression models in scenarios where the variables are highly correlated” and where there are many variables which is useful in this dataset as there is 19 columns, the actual vs predicted line in Fig 13 is on the line again which is a result of the data set its self and is the same as the baseline version mentioned previously.

Fig 13:

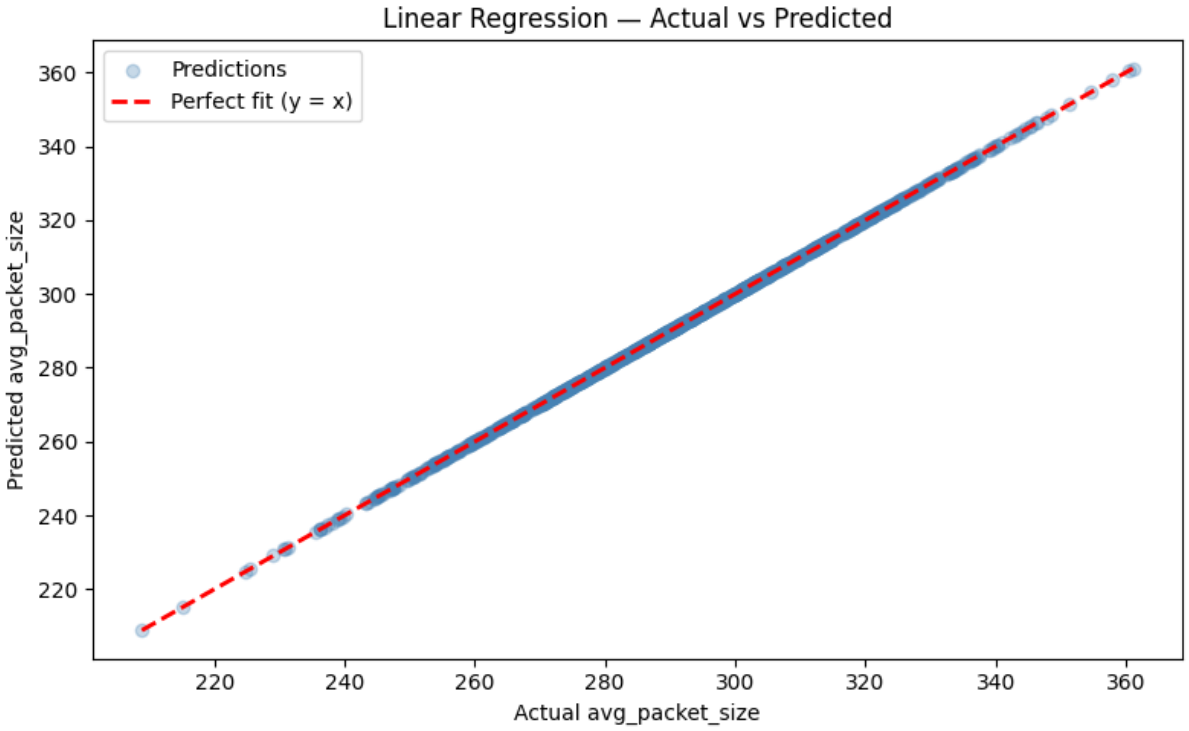
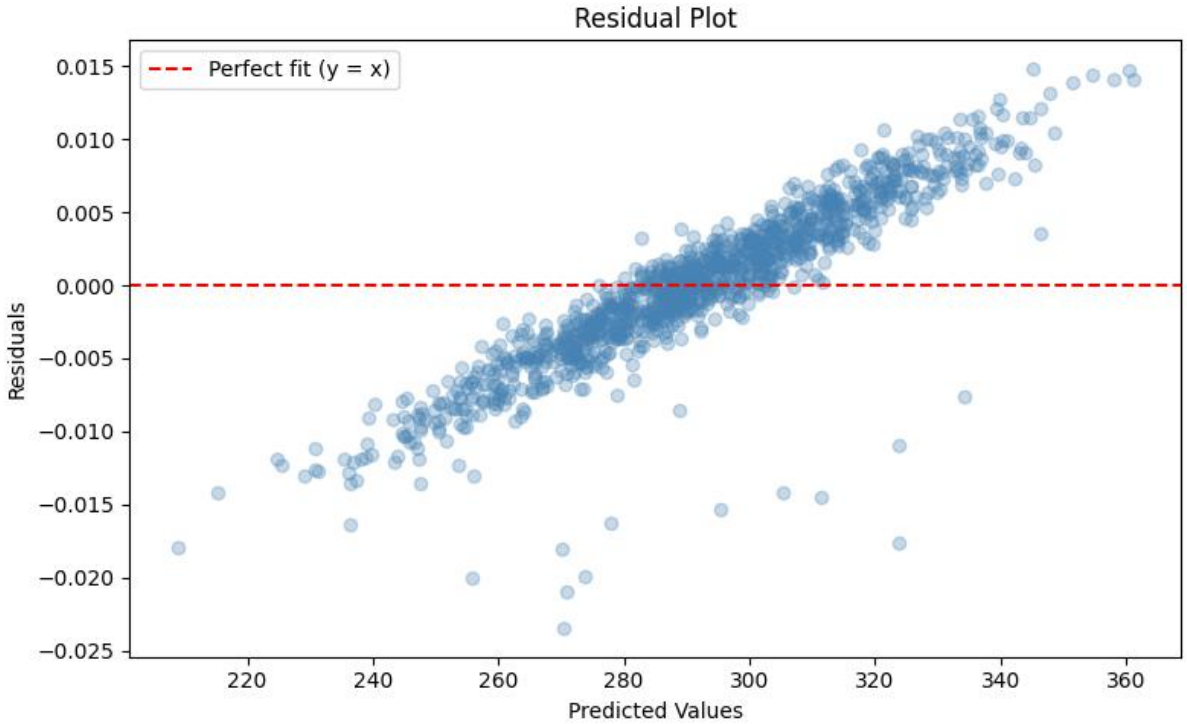



Fig 14:




The big difference comes from the residual plot, a residual plot "is a graphical representation of the residuals (errors) in a linear regression model" it shows the difference between the predicted and actual results and in the

case of the ridge it seems to be more realistic as the bigger the predicated the value the higher the residual so it can be called linear.

# Conclusion:

To conclude, given the dataset and using classification and predictive them being decision tree and linear regression, respectfully, and using the models to train the data set and give us the data and potential trends in easy to understand graphs and charts. From what I could make from the data is that the decision tree when tuned, makes a lot less mistakes such as false positives or negatives and the linear regression, is 100% accurate which shouldn't be possible in a real world dataset but possible in this one.

The dataset also shows that the majority of the data in the set is benign  around 60%-70% and from the brute-force attacks, there seems to be no region that its primarily coming from, it is evenly spread

# References:

Wikipedia. (2022). *Ridge regression*. [online] Available at: https://en.wikipedia.org/wiki/Ridge_regression.